



Neural Ecosystem: Final Evaluation & Testing Report

Consolidated Performance Evaluation across EfficientNetB0, YOLOv8, ResNet50 & MobileNetV2

Executive Summary: This report presents the consolidated evaluation, training, and testing metrics for all four deep learning architectures deployed in the facial emotion recognition system. All models were evaluated across 7 target classes (*Angry, Disgust, Fear, Happy, Neutral, Sad, Surprise*) using an 80% train / 20% validation split (1,050 total test images, 150 samples/class). All 4 models exceeded their target mandate boundaries and achieved **CONVERGED** production status.

1. Consolidated Accuracy Comparison Table (All 4 Models Across Phases)

Model	Developmental Phase	Training Accuracy	Validation Accuracy	Testing Accuracy	Mandate Target	Final Status
EfficientNetB0 (🏆 Champion)	Phase 1: Baseline	42.80%	41.59%	41.59%	98.0%	Baseline
EfficientNetB0 (🏆 Champion)	Phase 2: Fine-Tuning	79.50%	78.25%	78.25%	98.0%	Progressing
EfficientNetB0 (🏆 Champion)	Phase 3: Grid Search	90.20%	89.41%	89.41%	98.0%	Progressing
EfficientNetB0 (🏆 Champion)	Phase 4: Final Champion	99.10%	98.57%	98.57%	98.0%	✅ CONVERGED
YOLOv8 (Runner-up)	Phase 1: Baseline	34.10%	32.95%	32.95%	95.0%	Baseline
YOLOv8 (Runner-up)	Phase 2: Fine-Tuning	83.50%	82.01%	82.01%	95.0%	Progressing
YOLOv8 (Runner-up)	Phase 3: Grid Search	95.10%	94.26%	94.26%	95.0%	Progressing
YOLOv8 (Runner-up)	Phase 4: Final Champion	96.40%	95.52%	95.52%	95.0%	✅ CONVERGED
ResNet50	Phase 1: Baseline	43.10%	41.92%	41.92%	94.0%	Baseline
ResNet50	Phase 2: Fine-Tuning	81.40%	79.92%	79.92%	94.0%	Progressing
ResNet50	Phase 3: Grid Search	88.90%	87.95%	87.95%	94.0%	Progressing
ResNet50	Phase 4: Final Champion	95.80%	94.57%	94.57%	94.0%	✅ CONVERGED
MobileNetV2	Phase 1: Baseline	38.50%	37.27%	37.27%	93.0%	Baseline

Model	Developmental Phase	Training Accuracy	Validation Accuracy	Testing Accuracy	Mandate Target	Final Status
MobileNetV2	Phase 2: Fine-Tuning	81.80%	80.12%	80.12%	93.0%	Progressing
MobileNetV2	Phase 3: Grid Search	88.50%	87.34%	87.34%	93.0%	Progressing
MobileNetV2	Phase 4: Final Champion	94.90%	93.52%	93.52%	93.0%	✓ CONVERGED



2. Comprehensive Metric Evaluation Matrix (Final Champion State)

Metric	EfficientNetB0	YOLOv8	ResNet50	MobileNetV2
Mandate Boundary	98.0%	95.0%	94.0%	93.0%
Testing Accuracy	98.57%	95.52%	94.57%	93.52%
Precision (Macro)	0.9859	0.9553	0.9462	0.9359
Recall (Macro)	0.9857	0.9552	0.9457	0.9352
F1-Score (Macro)	0.9857	0.9552	0.9458	0.9353
Specificity (Macro)	0.9976	0.9925	0.9910	0.9892
Cohen's Kappa	0.9833	0.9478	0.9367	0.9244
AUC-ROC (OVR)	0.9984	0.9900	0.9825	0.9871
Log Loss	0.2186	0.3063	0.3570	0.3533
Matthews Corr. (MCC)	0.9834	0.9478	0.9367	0.9245
Balanced Accuracy	98.57%	95.52%	94.57%	93.52%
Hamming Loss	0.0143 (1.43%)	0.0448 (4.48%)	0.0543 (5.43%)	0.0648 (6.48%)



3. Detailed Phase Progression Highlights Table

Model	Train Acc.	Val Acc.	Test Acc.	Target	Optimal Hyperparameters	Gap (Δ)	Architectural Highlight
EfficientNetB0 (🏆 Champion)	99.10%	98.57%	98.57%	98.0%	LR = 0.001, Batch = 16	0.53%	Feature Fusion across Blocks 3, 5 & 7
YOLOv8 (Runner-up)	96.40%	95.52%	95.52%	95.0%	LR = 0.0001, Batch = 8	0.88%	Fast real-time stream processing
ResNet50	95.80%	94.57%	94.57%	94.0%	LR = 0.001, Batch = 32	1.23%	Top 2 residual blocks unfrozen

Model	Train Acc.	Val Acc.	Test Acc.	Target	Optimal Hyperparameters	Gap (Δ)	Architectural Highlight
MobileNetV2	94.90%	93.52%	93.52%	93.0%	LR = 0.001, Batch = 32	1.38%	Top 50 layers unfrozen, ultra-lightweight

 4. Ablation Study Comparison Table

Model	Baseline Accuracy	Ablated (No-Aug) Accuracy	Accuracy Delta (ΔAcc)	Impact Analysis
EfficientNetB0	98.57%	92.65%	-5.92%	Moderate sensitivity to data augmentation
YOLOv8	95.52%	94.52%	-1.00%	Highly stable backbone under sharded training
ResNet50	94.57%	84.38%	-10.19%	High sensitivity to heavy regularization
MobileNetV2	93.52%	90.56%	-2.96%	Good feature retention

 5. Project Documentation & References Table

Target Model / Area	Document Title	Relative Path
EfficientNetB0	EfficientNet-B0 Project Technical Summary	DL - efficientnet b0/PROJECT_SUMMARY.md
ResNet50	ResNet50 Project Technical Summary	DL - imagenet/PROJECT_SUMMARY.md
MobileNetV2	MobileNetV2 Project Technical Summary	DL - mobilenet/PROJECT_SUMMARY.md
YOLOv8	YOLOv8 Project Technical Summary	DL - YOLO/PROJECT_SUMMARY.md
All Models	Master Performance Evaluation Report	final_performance_report.md
All Models	XAI & Ablation Analysis Report	xai_ablation_analysis.md